NEET TEST HUB

NTH-PREMIUM TEST SERIES 06

Date – 26 October 2025 | **Duration – 3 Hours** | **Marks – 720**

Physics :- System of Particles & Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids

Chemistry :- Thermodynamics, Equilibrium, Redox Reactions

Biology :- Morphology of Flowering Plants, Anatomy of Flowering Plant, Breathing & Exchange of Gases, Body Fluids & Circulation, Excretory Products & their Elimination

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l.

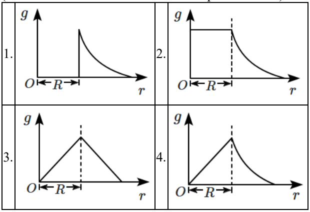
(A)	The loss in gravitational potential energy of M is Mgl.
(B)	The elastic potential energy stored in the wire is Mgl.
(C)	The elastic potential energy stored in the wire is $\frac{1}{2}Mgl$.
(D)	The heat produced is $\frac{1}{2}Mgl$.

Choose the correct option from the options given below:

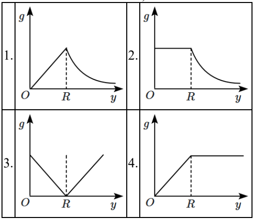
1.	**(A), (B)** and **(C)** only
2.	**(A)** and **(B)** only
3.	**(B)** and **(D)** only
4.	**(A), (C)** and **(D)** only

2. By sucking through a straw, a student can reduce the pressure in his lungs to 750 mm of Hg (density = 13.6 g/cm^3). Using the straw, he can drink water from a glass up to a maximum depth of:
1. 10 cm
2. 75 cm
3. 13.6 cm
4. 1.36 cm

3. Which one of the following plots represents the variation of a gravitational field on a particle with distance r due to a thin spherical shell of radius R? (r is measured from the centre of the spherical shell)



4. Which of the following graphs correctly represents the variation of acceleration due to gravity g as a function of the distance y from the centre of the Earth? (Here, R denotes the Earth's radius.)



5. n drops of a liquid, each with surface energy E, join to form a single drop.

(A)	Some energy will be released in the process.
(B)	Some energy will be absorbed in the process.
(C)	The energy released or absorbed will be $E(n-n^{2/3})$.
(D)	The energy released or absorbed will be $nE(2^{2/3}-1)$.

Choose the correct option from the options given below:

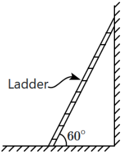
1.	**(A)** and **(C)** only
2.	**(A)** and **(B)** only
3.	**(B)** and **(D)** only
4.	**(B), (C)** and **(D)** only

6. The wheel of a motor accelerates uniformly from rest. In the first second, it rotates through an angle of 2 radians. What will be the angle through which it rotates in the next second?
1. 2 radians
2. 4 radians
3. 6 radians
4. 8 radians

7. Two wires are made of the same material and have the same volume. The first wire has a cross-sectional area A and the second wire has a cross-sectional area $3A$. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

1.	$9F$	2.	$6F$
3.	$4F$	4.	F

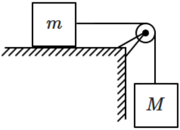
8. A ladder leaning against a wall makes an angle of 60° with the horizontal, as shown below. A person needs to climb this ladder.



The ladder is more likely to slip when the person:

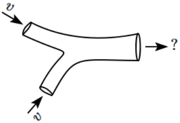
1.	just steps on the very bottom end of the ladder.
2.	stands on the ladder near the bottom end.
3.	stands on the ladder near the middle.
4.	stands on the ladder near the top end.

9. The figure shows two blocks of masses m and M connected by a string passing over a pulley. The horizontal table over which the mass m slides is smooth. The pulley has a radius r and moment of inertia I about its axis and it can freely rotate about this axis. The acceleration of the mass M assuming that the string does not slip on the pulley is:



1. $\frac{mg}{M+m+I/r^2}$
2. $\frac{Mg}{M+m+I/r^2}$
3. $\frac{Mg}{M+m+2I/r^2}$
4. $\frac{mg}{M+m+2I/r^2}$

10. Two veins of equal radii r each, both carrying blood flowing at the same speed v join together to form a single vein of twice the radius $(2r)$. The speed of blood flow in the combined vein is:



1.	v	2.	$\frac{v}{2}$
3.	$2v$	4.	$\frac{v}{4}$

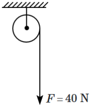
11. Air flows across the horizontal wings of an airplane in such a way that its speeds below and above the wings are 90 m/s and 120 m/s respectively. If the density of air is 1.3 kg/m^3, then the pressure difference across the lower and upper radii of the wings will be:

1.	7338 N/m^2	2.	3715 N/m^2
3.	3220 N/m^2	4.	4095 N/m^2

12. The energy required to shift a satellite from orbital radius r to orbital radius $2r$ is E. What energy will be required to shift the satellite from orbital radius $2r$ to orbital radius $3r$?
1. E
2. $E/2$
3. $E/3$
4. $E/4$

13. Three liquids of densities ρ_1, ρ_2 and ρ_3 $(\rho_1 > \rho_2 > \rho_3)$ having the same value of the surface tension T, rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey:
1. $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$
2. $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$
3. $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$
4. $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

14. The pulley has a radius of 50 cm and a moment of inertia of 0.5 kg-m^2. A frictional torque of 10 N-m acts on the axle. A constant force $F = 40$ N acts on the pulley through the light rope. The acceleration of the rope, as it unwinds, is (assume that the rope does not slip on the pulley)



1.	$10 \mathrm{~m/s^2}$	2.	$5 \mathrm{~m/s^2}$
3.	$2.5 \mathrm{~m/s^2}$	4.	$20 \mathrm{~m/s^2}$

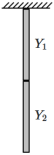
15. A liquid droplet of radius r breaks-up into N identical droplets. The surface area of system increases; the surface energy increases. The total surface energy of the smaller droplets increases with N as (r: fixed):
1. N
2. $N^{2/3}$
3. $N^{1/3}$
4. $N^{4/3}$

16. A wind with a speed of 40 m/s blows parallel to the roof of a house. The area of the roof is $250 \mathrm{~m}^2$. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be:
$(\rho_{\text{air}} = 1.2 \mathrm{~kg/m^3})$
1. 4×10^5 N, downwards
2. 4×10^5 N, upwards
3. 2.4×10^5 N, upwards
4. 2.4×10^5 N, downwards

17. An object of mass 50 g just floats in a liquid of density 2.5 g/ml. When the object is placed in a liquid of density 2.0 g/ml, it sinks to the bottom of the container. What is the force that the object exerts on the bottom of the container?
(take $g = 10 \mathrm{~m/s^2} = 10$ N/kg)

1.	0.1 N	2.	0.4 N
3.	10 N	4.	40 N

18. A combination of two wires of identical length (L, each) and cross-section (A, each) joined end-to-end behaves elastically as a single wire of Young's modulus Y: Y_1, Y_2 being the module of the individual wires. Then:



1. $Y = Y_1 + Y_2$
2. $Y = \frac{Y_1 + Y_2}{2}$
3. $\frac{1}{Y} = \frac{1}{Y_1} + \frac{1}{Y_2}$
4. $\frac{2}{Y} = \frac{1}{Y_1} + \frac{1}{Y_2}$

19. A flywheel, rotating about its axis, has a moment of inertia 0.16 kg-m^2. Its angular momentum decreases from 3.2 kg-m^2/s to 0.8 kg-m^2/s in 1.2 s. The magnitude of average torque acting on the flywheel is:
1. 2 N-m
2. 3.2 N-m
3. 1.6 N-m
4. 0.5 N-m

20. A uniform disc of mass m and radius r is rotating freely about its own axis, which is kept vertical. There is a small insect of mass $\frac{m}{2}$ sitting on the periphery of the disc, also rotating with the disc. The initial angular speed of the disc is ω_0. The insect moves radially inward and finally reaches the centre. The final angular speed of the disc is:

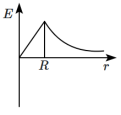
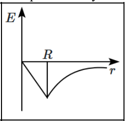
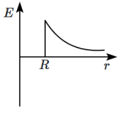
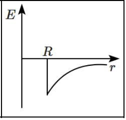
1.	ω_0	2.	$2\omega_0$
3.	$\frac{3}{2}\omega_0$	4.	$\frac{5}{2}\omega_0$

21. The moment of inertia of a thin uniform spherical shell and of a uniform solid sphere of the same masses and radii, taken about their respective diameters, are I_O (shell) and I_S (solid). Then:
1. $I_O = I_S$
2. $\frac{I_O}{2} = \frac{I_S}{3}$
3. $\frac{I_O}{3} = \frac{I_S}{5}$
4. $\frac{I_O}{5} = \frac{I_S}{3}$

22. Two stars of masses m and $2m$ at a distance d rotate about their common centre of mass in free space. The distance of the centre of mass from the heavier mass is:

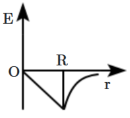
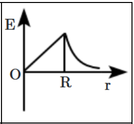
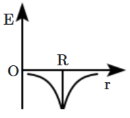
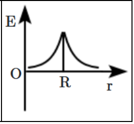
1.	$\frac{d}{3}$	2.	$\frac{2d}{3}$
3.	$\frac{4d}{3}$	4.	$\frac{5d}{3}$

23. The variation of intensity of the gravitational field (E) of the moon (having radius R) with distance (r) from the centre of the moon is represented by:

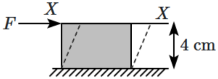
1.		2.	
3.		4.	

24. When a block of mass M is suspended by a long wire of length L, the length of the wire becomes $(L+l)$. The elastic potential energy stored in the extended wire is:
1. $\frac{1}{2}MgL$
2. Mgl
3. MgL
4. $\frac{1}{2}Mgl$

25. Dependence of intensity of gravitational field (E) of the earth with distance (r) from the centre of the earth is correctly represented by: (where R is the radius of the earth)

1.		2.	
3.		4.	

26. A steel plate of face area 1 cm^2 and thickness 4 cm is rigidly fixed at the lower surface. The tangential force $F = 10$ kN is applied on the upper surface, as shown in the figure. The lateral displacement x of the upper surface with respect to the lower surface is:
(given, the modulus of rigidly for steel is 8×10^{11} N/m^2)

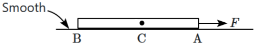


1. 5.0×10^{-5} m
2. 5.0×10^{-6} m
3. 2.5×10^{-3} m
4. 2.5×10^{-4} m

27. A new planet is discovered whose mass is the same as that of Earth, although the acceleration of freely falling objects at the surface of this planet is three times larger than that on Earth. (i.e., $g_{\text{Earth}} = 9.8$ m/s^2 and $g_{\text{planet}} = 29.4$ m/s^2).
If an object was dropped from a height h on this planet, how much time will it take to reach the ground compared to the time it would take on Earth to drop the same distance h? (assume no atmospheric resistance)

1.	It would take the same amount of time.
2.	The time would be less by a factor of 1.7.
3.	The time would be less by a factor of 3.
4.	The time would be less by a factor of 9.

28. Consider a uniform beam AB, which is being pulled by a horizontal force F applied at the end A, so that it is accelerated uniformly. The cross-section of the beam is A. Let the stress at the ends A, B be S_A, S_B and that at the centre C be S_C. Then:



1.	S_A = zero, S_B = maximum, S_C = intermediate
2.	S_A = maximum, $S_B = 0$, S_C = intermediate
3.	$S_A = S_B$ = maximum, S_C = zero
4.	$S_A = S_B = S_C$ = constant throughout

29. Assuming the earth to be a sphere of uniform density, its acceleration due to gravity acting on a body:

1.	increases with increasing altitude.
2.	increases with increasing depth.
3.	is independent of the mass of the earth.
4.	is independent of the mass of the body.

30. The dimensions of the ratio of pressure to the elastic modulus is:

1.	$[ML^{-2}]$	2.	$[LT^{-2}]$
3.	$[L^2]$	4.	$[M^0L^0T^0]$ (dimensionless)

31. Given below are two statements:

Statement I:	The kinetic energy of a planet is maximum when it is closest to the sun.
Statement II:	The time taken by a planet to move from the closest position (perihelion) to the farthest position (aphelion) is larger for a planet that is farther from the sun.

1.	**Statement I** is incorrect and **Statement II** is correct.
2.	Both **Statement I** and **Statement II** are correct.
3.	Both **Statement I** and **Statement II** are incorrect.
4.	**Statement I** is correct and **Statement II** is incorrect.

32. If the length of a wire is doubled and its radius is halved compared to their respective initial values, then, Young's modulus of the material of the wire will:

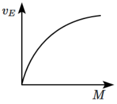
1.	remain the same.
2.	become 8 times its initial value.
3.	become $\frac{1}{4}$th of its initial value.
4.	become 4 times its initial value.

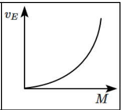
33. The angular velocity of a rotating body changes from ω_1 to ω_2 without the application of an external torque, but due to a change in its moment of inertia about the axis of rotation. The ratio of its corresponding radii of gyration is:
1. $\omega_1 : \omega_2$
2. $\sqrt{\omega_1} : \sqrt{\omega_2}$
3. $\omega_2 : \omega_1$
4. $\sqrt{\omega_2} : \sqrt{\omega_1}$

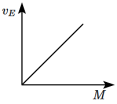
34. Water is under high pressure within a tank. When a hole is made at the top of the tank, a stream of water rises up vertically to a height H above the hole. Assume the flow to be streamlined. The water pressure (gauge pressure) at the top of the tank, initially, was: (ρ density of water)

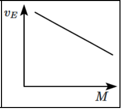
1.	$H\rho g$	2.	$\frac{H\rho g}{2}$
3.	$2H\rho g$	4.	$\sqrt{2}H\rho g$

35. Which graph represents the best possible variation of escape velocity (v_E) of earth with the mass of earth (M)? (the radius of the Earth is constant.)

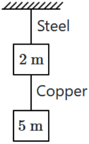
1. 

2. 

3. 

4. 

36. If the ratio of diameters, lengths and Young's modulus of steel and copper wires shown in the figure are p, q and s respectively, then the corresponding ratio of increase in their lengths would be:



1.	$\frac{5q}{(7sp^2)}$	2.	$\frac{7q}{(5sp^2)}$
3.	$\frac{2q}{(5sp)}$	4.	$\frac{7q}{(5sp)}$

37. The maximum elongation of a steel wire of $1\ \mathrm{m}$ length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8\ \mathrm{N\,m^{-2}}$ and $2 \times 10^{11}\ \mathrm{N\,m^{-2}}$, is:
1. 0.4 mm
2. 40 mm
3. 8 mm
4. 4 mm

38. If two soap bubbles of different radii are connected by a tube:

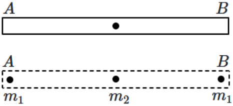
1.	air flows from the bigger bubble to the smaller bubble till the sizes become equal.
2.	air flows from the bigger bubble to the smaller bubble till the sizes are interchanged.
3.	air flows from the smaller bubble to the bigger.
4.	there is no flow of air.

39. The energy density in a wire which is under stress is given by:
1. $\frac{1}{2}(\text{stress})^2$
2. $\frac{1}{2}(\text{strain})^2$
3. $\frac{1}{2}(\text{stress}) \times (\text{strain})$
4. $\frac{1}{2}\frac{(\text{stress})^2}{\text{strain}}$

40. A uniform chain of length L is held so that its lowest point is in contact with the ground. The chain is released. When half of the chain has fallen, the centre-of-mass has descended by:

1.	$\frac{L}{2}$	2.	$\frac{L}{4}$
3.	$\frac{L}{8}$	4.	$\frac{3L}{8}$

41. A uniform rod AB of mass m and length L is replaced by three particles – two particles of masses m_1 each at the ends and another particle of mass m_2 at its centre. The new system of particles has the same total mass, the same center-of-mass and the same moment of inertia about an axis through its C.M. and perpendicular to AB.
Which of the following is true?



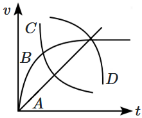
1.	$m_1 = \frac{m}{3}$, $m_2 = \frac{m}{3}$
2.	$m_1 = \frac{m}{4}$, $m_2 = \frac{m}{2}$
3.	$m_1 = \frac{m}{6}$, $m_2 = \frac{2m}{3}$
4.	$m_1 = \frac{m}{5}$, $m_2 = \frac{3m}{5}$

42. Given below are two statements:

Assertion (A):	When a body is raised from the surface of the earth, its potential energy increases.
Reason (R):	The potential energy of a body on the surface of the earth is zero.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

43. A spherical ball is dropped into a long column of a highly viscous liquid. The graph that represents the speed of the ball (v) as a function of time (t) is:



1.	D	2.	A
3.	B	4.	C

44. On any planet, the presence of atmosphere implies: (where v_{rms} = root mean square velocity of molecules and v_e = escape velocity)
1. $v_{rms} < v_e$
2. $v_{rms} > v_e$
3. $v_{rms} = v_e$
4. $v_{rms} = 0$

45. A projectile is fired with a constant speed v, at an angle θ above the horizontal. The angular momentum of the projectile about the point of projection, measured at the instant the projectile is at its highest point, is: (A : constant)
1. $A \sin\theta \cos\theta$
2. $A \sin^2\theta \cos\theta$
3. $A \sin\theta \cos^2\theta$
4. $A(\sin\theta + \cos\theta)$

Chemistry

46. Which of the following compounds has an element exhibiting two differrent oxidation states?

1.	NH_2OH	2.	NH_4NO_3
3.	N_2H_4	4.	N_3H

47. Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 liters of an ideal gas at 10 atmospheric pressure isothermally expands into a vacuum until the volume is 15 liters:

1.	Both the heat and work done will be greater than zero.
2.	Heat absorbed will be less than zero and work done will be positive.
3.	Work done will be zero and heat absorbed or evolved will also be zero.
4.	Work done will be greater than zero and heat absorbed will remain zero.

48. At 60°C, dinitrogen tetroxide is 50 percent dissociated. The standard free energy change at this temperature will be:

1. -863.8 $kJmol^{-1}$
2. -652.7 $kJmol^{-1}$
3. -763.8 $kJmol^{-1}$
4. -789.9 $kJmol^{-1}$

49. An example of homogeneous equilibrium among the following is:

1.	$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_3(g)$
2.	$C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$
3.	$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$
4.	$NH_4HS_{(s)} \rightleftharpoons NH_{3(s)} + H_2S_{(g)}$

50. Consider the following reaction at equilibrium at a temperature of T Kelvin, with a given equilibrium constant $K_c = 3 \times 10^{-13}$:

$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

Now, consider the reverse reaction:

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

The equilibrium constant for this reaction is denoted as $K_{c'}'$ which can be expressed as $a \times 10^{+b}$ in scientific notation.

What is the value of $a + b$?

1. 26
2. 29
3. 20
4. 19

51. From the following, identify the reaction having the top position in the EMF series (standard reduction potential) according to their electrode potential at 298 K.

1. $Mg^{2+} + 2e^{-} \rightarrow Mg_{(s)}$
2. $Fe^{2+} + 2e^{-} \rightarrow Fe_{(s)}$
3. $Au^{3+} + 3e^{-} \rightarrow Au_{(s)}$
4. $K^{+} + 1e^{-} \rightarrow K_{(s)}$

52. The value of ΔH for the reaction,

$X_2(g) + 4Y_2(g) \rightleftharpoons 2XY_4(g)$

is less than zero. Formulation of $XY_4(g)$ will be favored at:

1. Low pressure and Low temperature
2. High temperature and Low pressure
3. High pressure and Low temperature
4. High temperature and High pressure

53. Assuming water vapor to be a perfect gas, what is the change in internal energy when 1 mole of water(at 100°C and 1 bar pressure) is converted to ice at 0°C? (Given: Enthalpy of fusion of ice =6.0 kJ mol^{-1}; Heat capacity of water = 4.2 J/g°C)

1. -13.56 J mol^{-1}
2. -13.56 kJ mol^{-1}
3. -1.356 kJ mol^{-1}
4. -1.356 J mol^{-1}

54. What is the solubility product of $AuCl_3$, if the solubility of gold(III) chloride is 1.00×10^{-4} g L^{-1}? [Given: molar mass = 303.3 g/mol]

1.	1.00×10^{-16}	2.	2.7×10^{-15}
3.	1.2×10^{-26}	4.	3.2×10^{-25}

55. Which combination of the following substances will result in the formation of an acidic buffer when mixed?

1.	Weak acid and it's salt with a strong base.
2.	Equal volumes of equimolar solutions of weak acid and weak base.
3.	Strong acid and its salt with a strong base.
4.	Strong acid and its salt with a weak base. (The pK_a of acid = pK_b of the base)

56. MnO_4^{2-} (1 mole) in neutral aqueous medium will disproportionate to :
1. 2/3 Mole of MnO_4^- and 1/3 mole of MnO_2
2. 1/3 Mole of MnO_4^- and 2/3 mole of MnO_2
3. 1/3 Mole of Mn_2O_7 and 1/3 mole of MnO_2
4. 2/3 Mole of Mn_2O_7 and 1/3 mole of MnO_2

57. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A):	Pressure is an intensive property.
Reason (R):	Volume is an extensive property.

In the light of the above statements choose the correct answer from the options given below:

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

58. The oxidation number of the atom (in bold) in the following species is given. Identify, which one is incorrectly related?
1. $\mathbf{Cu}_2O$ is -1
2. $\mathbf{Cl}O_3^-$ is $+5$
3. $K_2\mathbf{Cr}_2O_7$ is $+6$
4. $H\mathbf{Au}Cl_4$ is $+3$

59. Which species are oxidized and reduced in the given reaction?

$FeC_2O_4 + KMnO_4 \rightarrow Fe^{3+} + CO_2 + Mn^{2+}$

1. Oxidised : Fe,C ; Reduced : Mn
2. Oxidised : Fe; Reduced : Mn
3. Reduced : Fe,Mn ; Oxidised : C
4. Reduced : C; Oxidised : Mn,Fe

60. The solubility product of a sparingly soluble salt AX_2 is 3.2×10^{-11}. Its solubility (in mol L^{-1}) is:
1. 5.6×10^{-6}
2. 3.1×10^{-4}
3. 2×10^{-4}
4. 4×10^{-4}

61. The oxidation number of potassium in K_2O, K_2O_2 and KO_2, respectively, is:
1. $+2, +1$ and $+\frac{1}{2}$
2. $+1, +1$ and $+1$
3. $+1, +4$ and $+2$
4. $+1, +2$ and $+4$

62. $$K_{eq} = \frac{(P_{SO_2})^2 \cdot P_{O_2}}{(P_{SO_3})^2}$$

Which of the reactions listed below match the equilibrium expression given above?
1. $2\ SO_2\ (aq) + O_2\ (g) \rightleftharpoons 2\ SO_3\ (aq)$
2. $2\ SO_3\ (aq) \rightleftharpoons 2\ SO_2\ (aq) + O_2\ (g)$
3. $2\ SO_2\ (g) + O_2\ (g) \rightleftharpoons 2SO_3\ (g)$
4. $2\ SO_3\ (g) \rightleftharpoons 2\ SO_2\ (g) + O_2\ (g)$

63. If the equilibrium constant for $N_2(g) + O_2\ (g) \rightleftarrows 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftarrows NO(g)$ will be?
1. $K^{\frac{1}{2}}$
2. $\frac{1}{2}K$
3. K
4. K^2

64. Fluorine is a strong oxidising agent because:
1. it has several isotopes
2. it is very small and has 7 electrons in valency shell
3. its valency is one
4. it is the first member of the halogen series

65. What is the approximate pH of a basic buffer solution that contains 0.0025 moles of ammonium chloride (NH_4Cl) and 0.15 moles of ammonium hydroxide (NH_4OH)?
[Given: pKb of ammonium hydroxide = 4.74]

1.	11.04	2.	10.24
3.	6.52	4.	5.48

66. Hydrolysis of sucrose is given by the following reaction

Sucrose + $H_2O \rightleftharpoons$ Glucose + Fructose

If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^\ominus$ at the same temperature will be:

1. 8.314 J mol^{-1} $K^{-1}\times$300 K$\times$ln (2×10^{13})
2. 8.314 J mol^{-1} $K^{-1}\times$300 K$\times$ln (3×10^{13})
3. –8.314 J mol^{-1} $K^{-1}\times$300 K$\times$ln (4×10^{13})
4. –8.314 J mol^{-1} $K^{-1}\times$300 K$\times$ln (2×10^{13})

67. One mole of $N_2(g)$ is mixed with 2 moles of $H_2(g)$ in a 4 litre vessel. 50% of $N_2(g)$ is converted to $NH_3(g)$ by the following reaction:

$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$

What will be the value of K_c for the following equilibrium?

$NH_3\left(g\right) \rightleftharpoons \frac{1}{2}N_2\left(g\right) + \frac{3}{2}H_2\left(g\right)$

1. 256
2. 16
3. $\frac{1}{16}$
4. None of the above

68. If a gas absorbs 200 J of heat and expands by 500 cm^3 against the constant pressure of $2x10^5$ Nm^{-2}, then the change in internal energy is :

1. -200 J
2. -100J
3. +100J
4. + 300J

69. Given below are two statements:

Statement I:	$KMnO_4$ acts as a self-indicator.
Statement II:	$K_2Cr_2O_7$ is not used as a self-indicator.

Choose the correct option:

1.	**Statement I** is correct but **Statement II** is incorrect.
2.	**Statement I** and **Statement II** both are correct.
3.	**Statement I** and **Statement II** both are incorrect.
4.	**Statement I** is incorrect but **Statement II** is correct.

70. Match the following reactions in **Column-I** to the correct enthalpy associated with them given in **Column-II**:

	Column-I		Column-II
A.	$C_6H_6(l) + \frac{15}{2}O_2 \rightarrow 6CO_2(g) + 3H_2O(l)$	I.	$\Delta_{\text{lattice}}H^\circ$
B.	$NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$	II.	$\Delta_{\text{hyd}}H^\circ$
C.	$AB(s) \rightarrow A^+(aq) + B^-(aq)$	III.	$\Delta_c H^\circ$
D.	$A^+(g) + B^-(g) \rightarrow A^+(aq) + B^-(aq)$	IV.	$\Delta_{\text{sol}}H^\circ$

Choose the correct option from the following:

1. A - (III); B - (I); C - (IV); D - (II)
2. A - (III); B - (I); C - (II); D - (IV)
3. A - (II); B - (I); C - (IV); D - (III)
4. A - (III); B - (II); C - (I); D - (IV)

71. Arrange the metals Al, Cu, Fe, Mg, and Zn in the correct decreasing order based on their ability to displace each other from the solution of their salts:

1. Al> Zn > Fe > Cu > Mg
2. Zn > Fe > Cu > Mg > Al
3. Mg > Al > Zn > Fe > Cu
4. Fe > Mg > Zn > Al > Cu

72. Which of the following cases leads to the reaction remaining fastest to completion?

1. $K = 1$
2. $K = 10$
3. $K = 10^{-2}$
4. $K = 10^2$

73. Which of the following reactions has/have a positive ΔS (entropy change)?

(i) Ag^+ (aq) + Cl^-(aq) $\rightarrow$AgCl(s)
(ii) $NH_4Cl(g) \rightarrow NH_3(g) + HCl(g)$
(iii) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$

1. (i) & (ii)
2. (iii)
3. (ii) & (iii)
4. (ii)

74. What is the oxidation number of Cl in $CaOCl_2$ (bleaching powder)?

1. Zero, because it contains Cl_2
2. -1, because it contains Cl^-
3. +1, because it contains ClO^-
4. +1 and -1 because it contains ClO^- and Cl^-

75. At 300K, ΔrG and ΔrG° are - 12.8 KJ mol^{-1} and -11.5 KJ mol^{-1} respectively for the following reaction:
$A(g) + B(g) \rightleftharpoons C(l) + D(aq)$
If the reaction is at equilibrium, then the equilibrium constant K will be:
1. -2.0×10^{0}
2. 1.0×10^{2}
3. 1.0×10^{-2}
4. 1.69×10^{0}

76. The oxidation numbers of sulphur in S_8, S_2F_2, H_2S respectively, are:
1. 0, +1 and -2
2. +2, +1 and -2
3. 0, +1 and +2
4. -2, +1 and -2

77. Given that the total pressure inside a reaction vessel is 1.12 atm at 105°C for the reaction $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$.
What is the value of K_p?
1. 0.56
2. 1.25
3. 0.31
4. 0.63

78.

Assertion (A):	In the reaction, $H_2\ (g) +\ I_2\ (g) \rightleftharpoons 2HI\ (g)$, equilibrium constants K_p and K_c are equal.
Reason (R):	K_p of all gaseous reactions is equal to K_c.

In light of the above statements choose answer from the options given below:

1.	Both (**A**) and (**R**) are True and (**R**) is the correct explanation of (**A**).
2.	Both (**A**) and (**R**) are True but (**R**) is not the correct explanation of (**A**).
3.	(**A**) is True but (**R**) is False
4.	Both (**A**) and (**R**) are False.

79. An invalid equation among the following is:
1. $\Delta H = \Sigma H_{products} - \Sigma H_{reactants}$
2. $\Delta H = \Delta U + P\Delta V$
3. $\Delta H^{o}_{(reaction)} = \Delta H^{o}_{(products\ bonds)} - \Delta H^{o}_{(reactant\ bonds)}$
4. $\Delta H = \Delta U + \Delta nRT$

80. The oxidation number of carbon in CH_2O is:

1.	–2	2.	+2
3.	0	4.	+4

81. Which reaction is **NOT** a redox reaction?
1. $2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$
2. $H_2 + Cl_2 \rightarrow 2HCl$
3. $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$
4. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

82. The value $|\Delta H|$ in kJ for the given reaction is:
$\frac{1}{2}Cl_2(g) \rightarrow Cl^-(aq)$
(Given: $\Delta H_{\text{diss}}\ Cl_2(g) \rightarrow 2Cl(g)$ 240 $kJmol^{-1}$
$\Delta H_{eg} Cl(g) + e^- \rightarrow Cl^-(g) - 320\ kJmol^{-1}$
$\Delta H_{hydration} Cl^-(g) + aq \rightarrow Cl^-(aq) - 340\ kJmol^{-1}$)

1.	540	2.	620
3.	450	4.	470

83. Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is:
$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_{6(l)}$
[Given:
$\Delta_f G^{\circ}\left(HC \equiv CH\right) = -2.04 \times 10^5\ J\ mol^{-1}$
$\Delta_f G^{\circ}\ (C_6H_6) = -1.24 \times 10^5\ J\ mol^{-1}$
$R = 8.314\ J\ K^{-1}\ mol^{-1}$]
1. 860
2. 875
3. 855
4. 895

84. Match the terms in List I with their corresponding descriptions in List II:

	List I (Term)		**List II** (Description)
A.	Adiabatic process	i.	At constant temperature
B.	Isolated system	ii.	No transfer of heat
C.	Isothermal change	iii.	Heat
D.	Path function	iv.	No exchange of energy and matter

Codes:

	A	B	C	D
1.	ii	iv	i	iii
2.	iii	iv	i	ii
3.	iv	iii	i	ii
4.	iv	ii	i	iii

85. Which species cannot act as both an oxidant and a reductant?
1. H_2O_2
2. SO_2
3. SO_3
4. HNO_2

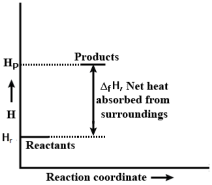
86. Consider the following reduction processes:
$Zn^{2+} + 2e^- \rightarrow Zn(s); E^\circ = -0.76\ V$
$Ca^{2+} + 2e^- \rightarrow Ca(s); E^\circ = -2.87\ V$
$Mg^{2+} + 2e^- \rightarrow Mg(s); E^\circ = -2.36\ V$
$Ni^{2+} + 2e^- \rightarrow Ni(s); E^\circ = -0.25\ V$
What is the ascending order of reducing power for the given metals?
1. Ca < Zn < Mg < Ni
2. Ni < Zn < Mg < Ca
3. Zn < Mg < Ni < Ca
4. Ca < Mg < Zn < Ni

87. Match the salt given in **Column-I** with the corresponding pH given in **Column-II** and mark the appropriate choice.

	Column-I		**Column-II**
(A)	CH_3COONa	(i)	Almost neutral, pH > 7 or <7
(B)	NH_4Cl	(ii)	Acidic, pH <7
(C)	$NaNO_3$	(iii)	Alkaline, pH >7
(D)	CH_3COONH_4	(iv)	Neutral, pH=7

1. (A) →(i), (B)→(ii), (C) →(iii), (D)→ (iv)
2. (A) →(ii), (B)→(iii), (C) →(iv), (D)→ (i)
3. (A) →(iii), (B)→(ii), (C) →(iv), (D)→ (i)
4. (A) →(iv), (B)→(i), (C) →(iii), (D)→ (ii)

88. An enthalpy diagram for a particular reaction is given below:



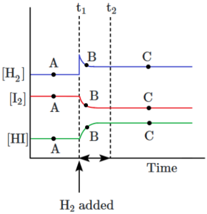
Which of the following statements is correct?

1.	Reaction is spontaneous
2.	Reaction is non-spontaneous
3.	The spontaneity of the reaction cannot be determined from the graph provided above.
4.	None of the above

89. Which one of the following reactions obeys $\Delta H \neq \Delta E$.?

1.	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
2.	$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
3.	$C(s) + O_2(g) \rightleftharpoons CO_2(g)$
4.	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

90. In the reaction $H_2 + I_2 \rightleftharpoons 2HI$, H_2 has been added at equilibrium, as depicted in the following graph:



The point that represents the re-established equilibrium is:
1. A
2. B
3. C
4. None of the above.

Biology

91. All the following would be true for a typical leaf in flowering plants except:

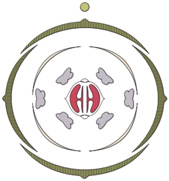
1.	The leaf develops at the node and bears a bud in its axil.
2.	Leaves originate from shoot apical meristems.
3.	Leaves are arranged in a basipetal order.
4.	Leaves are the most important vegetative organs for photosynthesis.

92. Match List I with List II:

	List I		List II
A.	P wave	I.	Heart muscles are electrically silent
B.	QRS complex	II.	Depolarisation of ventricles.
C.	T wave	III.	Depolarisation of atria.
D.	T-P gap	IV.	Repolarisation of ventricles.

Choose the correct answer from the options given below:
1. A-III, B-II, C-IV, D-I
2. A-II, B-III, C-I, D-IV
3. A-IV, B-II, C-I, D-III
4. A-I, B-III, C-IV, D-II

93. The Floral Diagram represents which one of the following families?



1.	Fabaceae	2.	Brassicaceae
3.	Solanaceae	4.	Liliaceae

94. Consider the given two statements:

Assertion(A):	Blood flows through the human heart in a single, continuous circuit.
Reason(R):	The human heart has four chambers - two auricles and two ventricles.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

95. Rh incompatibility can lead to grave consequence in cases where:

1.	an Rh negative mother is carrying an Rh positive foetus
2.	an Rh negative mother is carrying an Rh negative foetus
3.	an Rh positive mother is carrying an Rh positive foetus
4.	an Rh positive mother is carrying an Rh negative foetus

96. The descending limb of Henle's loop:

1.	Is permeable to water but impermeable to salts.
2.	Is impermeable to water but permeable to salts.
3.	Actively secretes sodium into the filtrate.
4.	Actively reabsorbs glucose and amino acids.

97. Why is the juxta-glomerular apparatus (JGA) crucial for regulating kidney function?

1.	It senses changes in filtrate volume in the proximal convoluted tubule.
2.	It facilitates the reabsorption of Na+ and Cl- ions in the nephron.
3.	It regulates the permeability of the Henle's loop to electrolytes.
4.	It releases renin in response to a drop in GFR, which triggers a cascade to increase blood flow to the glomerulus.

98. Regarding the regulation of respiration by brain:

I:	The rhythm center is located in the medulla oblongata.
II:	Pneumotaxic center is located in the mid brain

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is incorrect; **Statement II** is correct
3.	**Statement I** is correct; **Statement II** is incorrect
4.	**Statement I** is incorrect; **Statement II** is incorrect

99. Match each item in **Column-I** with one item in **Column-II**.

	Column-I		Column-II
A.	Twisted aestivation	P.	Pea
B.	Valvate aestivation	Q.	Cassia
C.	Vexillary aestivation	R.	China rose
D.	Imbricate aestivation	S.	*Calotropis*

Options	A	B	C	D
1	R	S	P	Q
2	Q	R	S	P
3	S	Q	R	P
4	P	Q	S	R

100. A person with an unknown blood group under the ABO system, has suffered significant blood loss in an accident and needs an immediate blood transfusion. His friend who has a valid certificate of his blood type, offers blood donation without delay. What would have been the type of blood group of the donor friend?
1. Type AB
2. Type O
3. Type A
4. Type B

101. Which of the following statements are correct with respect to vital capacity?

(a)	It includes ERV, TV and IRV
(b)	Total volume of air a person can inspire after a normal expiration
(c)	The maximum volume of air a person can breathe in after forced expiration
(d)	It includes ERV, RV and IRV.
(e)	The maximum volume of air a person can breathe out after a forced inspiration.

Choose the most appropriate answer from the options given below:

1.	(b), (d) and (e)	2.	(a), (c) and (d)
3.	(a), (c) and (e)	4.	(a) and (e)

102. Which of the following correctly explains the ground tissue system in dicot leaves?

1.	Composed exclusively of sclerenchyma and collenchyma for structural support.
2.	Composed of mesophyll tissue containing palisade parenchyma and spongy parenchyma, specialized for photosynthesis.
3.	Found only in the stem and root cortex, aiding in nutrient storage and transport.
4.	Contains only thin-walled parenchyma in the vascular bundles for conduction of water and food.

103. Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme:
1. thrombokinase
2. throboplastin
3. thrombin
4. streptokinase

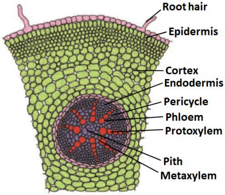
104. The respiratory rhythm centre is located in:
1. Cerebrum.
2. Cerebellum.
3. Pons.
4. Medulla.

105. Which part of the brain regulates the rhythmicity of breathing?
1. Cerebrum
2. Medulla oblongata
3. Hypothalamus
4. Pons

106. Which of the following statements correctly explains the role of guard cells in the epidermal tissue system?

1.	Guard cells regulate gaseous exchange but do not contain chloroplasts, so they are unable to photosynthesize.
2.	Guard cells have thickened outer walls and thin inner walls, allowing for efficient stomatal closure.
3.	Guard cells possess chloroplasts and control the opening and closing of stomata by regulating turgor pressure.
4.	Guard cells are specialized epidermal cells that only assist in preventing excessive water loss.

107. The given figure shows the T.S. of a:

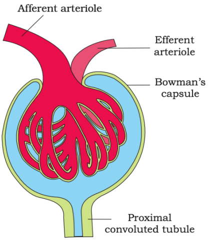


1. Monocot root
2. Dicot root
3. Dicot stem
4. Monocot stem

108. In the regulation of cardiac activity, the parasympathetic nervous system:

1.	Increases heart rate and the strength of ventricular contraction.
2.	Releases norepinephrine to increase cardiac output.
3.	Causes dilation of coronary arteries to increase oxygen delivery.
4.	Decreases heart rate and slows down conduction of action potentials.

109. The structure shown in the given figure:



1.	Is called as the Malpighian tubule
2.	Is the site of ultrafiltration
3.	Represents juxta-glomerular apparatus
4.	Is responsible for secretion of ANF

110. The gritty texture of pears is caused by

1.	sclereids	2.	collenchyma
3.	tracheids	4.	parenchyma

111. The release of Atrial Natriuretic Factor (ANF) by the heart acts as a check on the renin-angiotensin mechanism by:

1.	Promoting vasoconstriction to increase blood pressure.
2.	Increasing sodium reabsorption in the distal tubule.
3.	Increasing ADH release to promote water retention.
4.	Promoting vasodilation and decreasing blood pressure.

112. Following are the stages of pathway for conduction of an action potential through the heart:
A. AV bundle
B. Purkinje fibres
C. AV node
D. Bundle branches
E. SA node
Choose the correct sequence of pathway from the options given below :
1. A-E-C-B-D
2. B-D-E-C-A
3. E-A-D-B-C
4. E-C-A-D-B

113. The transverse section of monocot root shows the following internal tissue organization. Arrange them in correct sequence starting from periphery to the centre.
A. Endodermis
B. Pith
C. Epidermis
D. Pericycle
E. Cortex
Choose the correct answer from the options given below:
1. D, C, E, A, B
2. A, C, E, B, D
3. C, E, A, D, B
4. C, E, D, B, A

114. Which of the following statements about white blood cells (leukocytes) is NOT correct?

1.	Neutrophils are the most abundant type of white blood cells in most mammals and form an essential part of the innate immune system.
2.	Lymphocytes are involved in the creation of antibodies, which are crucial for the adaptive immune response.
3.	Monocytes mature into macrophages and dendritic cells, which can engulf and destroy pathogens through phagocytosis.
4.	Eosinophils are primarily responsible for the body's response to parasitic infections and are the predominant white blood cell type in human blood

115. The role of other organs in excretion includes the skin, which:
1. Excretes excess water and salts.
2. Produces bile to aid in digestion.
3. Filters blood to remove urea.
4. Secretes hormones to regulate blood pressure.

116. When compared with carbon dioxide, the partial pressure of oxygen in the air and its solubility in water is respectively:

1.	lower, lower	2.	lower, higher
3.	greater, lower	4.	greater, higher

117. The most common renal stones are:
1. oxalates
2. urates
3. cysteine
4. struvites

118. In humans, the process of respiration involves the following steps:

A.	Diffusion of gases across alveolar membrane
B.	Diffusion of gases between blood and tissues
C.	Transport of gases by blood
D.	Utilisation of O_2 by the cells for catabolic reactions
E.	Breathing or pulmonary ventilation

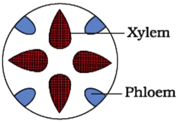
Choose the correct sequence of steps from the options given below:

1.	A →E→B→C→D	2.	C →E→A→B→D
3.	B →D→C→E→A	4.	E →A→C→B→D

119. Name the blood cells, whose reduction in number can cause clotting disorder, leading to excessive loss of blood from the body:
1. Erythrocytes
2. Leucocytes
3. Neutrophils
4. Thrombocytes

120. Which hormone directly causes the reabsorption of Na+ and water from the distal parts of the tubule?
1. ADH
2. ANF
3. Angiotensin II
4. Aldosterone

121. The following diagram depicts the type of vascular bundles found in:



A:	Monocot Roots
B:	Monocot Stems
C:	Dicot Roots
D:	Dicot Stems

1. Only **A** and **B**
2. Only **B** and **C**
3. Only **A** and **C**
4. Only **B** and **D**

122. Consider the following statements about human blood:

I:	Red blood cells in humans have a nucleus.
II:	Blood plasma contains fibrinogen which helps in clotting.

1. Only **Statement I** is correct.
2. Only **Statement II** is correct.
3. Both **Statements** are correct.
4. Neither **Statement** is correct.

123. When the heart muscle is suddenly damaged by an inadequate blood supply, the condition is called as

1.	Heart Attack	2.	Heart Failure
3.	Cardiac Arrest	4.	Angina pectoris

124. Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I		Column II
A.	Monocot stem	P.	Conjoint and closed vascular bundles
B.	Monocot root	Q.	Polyarch xylem, radial vascular bundles
C.	Dicot stem	R.	Conjoint and open vascular bundles
D.	Dicot root	S.	Two to four xylem and phloem patches

Codes:

	A	B	C	D
1.	P	Q	R	S
2.	S	R	Q	P
3.	P	Q	S	R
4.	S	R	P	Q

125. During the concentration of urine by human kidneys:

I:	Small amounts of urea is transported back to medullary interstitium by ascending portion of vasa recta
II:	NaCl is returned to the medullary interstitium by the descending portion of vasa recta

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

126. In the process of breathing, inspiration occurs when:

1.	The intra-pulmonary pressure is higher than atmospheric pressure.
2.	The intra-pulmonary pressure is less than atmospheric pressure.
3.	The pulmonary volume decreases.
4.	The diaphragm relaxes.

127. Match **Column-I** with values given in **Column-II**:

	Column-I Partial Pressure of		**Column-II** Value (in mm of Hg)
A.	Carbon dioxide in tissues	1.	104
B.	Carbon dioxide in alveoli	2.	40
C.	Oxygen in alveoli	3.	45
D.	Oxygen in oxygenated blood	4.	95

1. A-1, B-2, C-4, D-3
2. A-3, B-1, C-2, D-4
3. A-3, B-2, C-1, D-4
4. A-2, B-1, C-3, D-4

128. The fleshy leaves of plants like onion and garlic are modified to perform:
1. Food storage
2. Respiration
3. Climbing
4. Protection from herbivores

129. In China Rose:
A: Aestivation is called as twisted
B: Stamens are monadelphous
C: Placentation is axile.
1. Only **A** and **B** are correct.
2. Only **A** and **C** are correct.
3. Only **B** and **C** are correct.
4. **A**, **B**, and **C** are correct.

130. Which one of the following statements is correct?

1.	The seed in grasses is not endospermic
2.	Mango is a parthenocarpic fruit
3.	A proteinaceous aleurone layer is present in maize grain.
4.	A sterile pistil is called a staminode.

131. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the:

1.	mother between 20 weeks and 24 weeks of pregnancy
2.	mother early in pregnancy
3.	mother immediately after the delivery of the child
4.	new-born immediately after the delivery

132. If an Rh –ve mother is carrying an Rh +ve foetus, she is likely to be exposed to the Rh antigen:
1. At the time of implantation of the blastocyst
2. At the time when the placenta becomes functional
3. When the first movements of the baby can be felt
4. At the time of delivery

133. The amount of carbon dioxide delivered by 100 ml of deoxygenated blood at alveoli under normal physiological conditions:

1.	is equal to the amount of oxygen delivered by 100 ml of oxygenated blood at tissues under normal physiological conditions.
2.	is less than the amount of oxygen delivered by 100 ml of oxygenated blood at tissues under normal physiological conditions.
3.	is twice the amount of oxygen delivered by 100 ml of oxygenated blood at tissues under normal physiological conditions.
4.	is more than 10 times higher than the amount of oxygen delivered by 100 ml of oxygenated blood at tissues under normal physiological conditions.

134. Removal of the proximal convoluted tubule from the nephron will result in:
1. more diluted urine
2. more concentrated urine
3. no change in the quality and quantity of urine
4. no urine formation

135. The number of correct statements from the given statements is:

I:	In a typical dicotyledonous embryo, the portion of the embryonal axis above the level of cotyledons is the epicotyl, which terminates with the plumule or stem tip.
II:	In a typical dicotyledonous embryo, the cylindrical portion below the level of cotyledons is hypocotyl, and it terminates at its lower end in the radicle or root tip.
III:	In the grass family, the cotyledon is called the scutellum, which is situated towards one side (lateral) of the embryonal axis.
IV:	At its lower end, the embryonal axis in monocots has the radical and root cap enclosed in an undifferentiated sheath called coleorhiza.
V:	In monocots, the portion of the embryonal axis above the level of attachment of the scutellum is the epicotyl.

1.	2	2.	3
3.	4	4.	5

136. How many of the given statements are true?

A:	Alternate phyllotaxy is seen in China Rose, Mustard, and Sunflower.
B:	Opposite phyllotaxy is seen in Calotropis and Guava.
C:	Whorled phyllotaxy is seen in Alstonia.

Mark the correct option:
1. 0
2. 1
3. 2
4. 3

137. Consider the given two statements:

Assertion (A):	The parasympathetic nervous system increases cardiac output.
Reason (R):	The parasympathetic nervous system increases the heart rate and the strength of the heartbeat.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is False, **(R)** is False.
4.	**(A)** is False, **(R)** is True.

138. The main function of lenticel is:

1.	Transpiration	2.	Guttation
3.	Gaseous exchange	4.	Bleeding

139. Which of the following helps in maintenance of the pressure gradient in sieve tubes?
1. Albuminous cells
2. Sieve cells
3. Phloem parenchyma
4. Companion cells

140. Consider the given two statements:

Statement I:	The Renin-Angiotensin mechanism is triggered by a decrease in glomerular blood pressure or blood flow.
Statement II:	Angiotensin II acts as a vasoconstrictor and stimulates aldosterone release, increasing sodium and water reabsorption, thus raising blood pressure.

Options:

1.	Both **Statement I** and **Statement II** are correct.
2.	**Statement I** is correct, but **Statement II** is incorrect.
3.	**Statement I** is incorrect, but **Statement II** is correct.
4.	Both **Statement I** and **Statement II** are incorrect.

141. The given Figure 1 shows a floral diagram and Figure 2 shows a floral formula. Identify the correct statement with respect to the given figures:

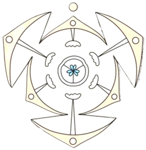


Figure - 1

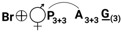


Figure - 2

1.	The floral diagram is for Family Fabacea and the floral formula is for Family Solanaceae
2.	The floral diagram is for Family Solaneceae and the floral formula is for Family Fabaceae
3.	Both floral diagram and floral formula are for Family Liliaceae
4.	Both floral diagram and floral formula are for Family Solanaceae

142. Inhalation of air into the lungs can be brought about by:

I:	Contraction of diaphragm
II:	Contraction of internal intercostal muscles
III:	Contraction of external intercostal muscles

1. Only **I** and **II**
2. Only **II**
3. Only **I** and **III**
4. Only **III**

143. Which of the following factors does not favor the release of oxygen from hemoglobin at the tissues?
1. High partial pressure of CO_2 (pCO_2).
2. Low pH (acidic conditions).
3. High temperature.
4. High partial pressure of O_2 (pO_2).

144. Which of the following would help in prevention of diuresis?

1.	Reabsorption of Na^+ and water from renal tubules due to aldosterone
2.	Atrial natriuretic factor causes vasoconstriction
3.	Decrease in the secretion of renin by JG cells
4.	More water reabsorption due to under secretion of ADH

145. Site of Ultra-filtration of the blood in the nephron is the:

1.	PCT	2.	DCT
3.	Collecting ducts	4.	Malpighian body

146. In dicot stems, each vascular bundle is:

1.	conjoint, open, and with endarch protoxylem.
2.	conjoint, closed, and with endarch protoxylem.
3.	radial, open, and with endarch protoxylem.
4.	conjoint, open, and with exarch protoxylem.

147. Consider the two statements:

Statement I:	People who are blood type O have both anti-A and anti-B antibodies present in their plasma.
Statement II:	People who are blood type AB do not produce either anti-A or anti-B antibodies.

1. **Statement I** is correct: **Statement II** is incorrect
2. **Statement I** is incorrect: **Statement II** is correct
3. **Statement I** is correct: **Statement II** is correct
4. **Statement I** is incorrect: **Statement II** is incorrect

148. A terrestrial organism must
1. conserve water
2. have a large body size
3. Have a separate genitary and urinary system
4. store a large amount of water in the alimentary canal

149. Which of the following vascular bundle arrangements is unique to roots, and how does it function?

1.	Radial arrangement with alternating xylem and phloem on different radii, allowing efficient water and food conduction.
2.	Conjoint arrangement with xylem and phloem on the same radius, supporting secondary growth.
3.	Closed bundles with no cambium, enabling rapid elongation of primary roots.
4.	Vascular bundles with only xylem elements specialized for water conduction in monocots.

150. Terrestrial organisms, like mammals, adapted to excrete urea instead of ammonia primarily because:

1.	Ammonia can be stored for long periods in the body without toxicity.
2.	Urea requires less water for excretion compared to ammonia, conserving water.
3.	Urea can be excreted directly from the skin without using the kidneys.
4.	Urea is more toxic than ammonia and requires less energy to produce.

151. Identify the correct statements:

A:	Lenticels are the lens-shaped openings permitting the exchange of gases.
B:	Bark formed early in the season is called hard bark.
C:	Bark is a technical term that refers to all tissues exterior to vascular cambium.
D:	Bark refers to periderm and secondary phloem.
E:	Phellogen is single- layered in thickness.

Choose the correct answer from the options given below:

1.	**B** and **C** only	2.	**B**, **C** and **E** only
3.	**A** and **D** only	4.	**A**, **B** and **D** only

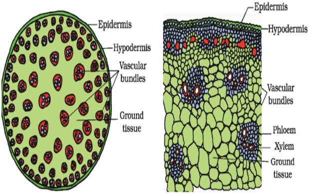
152. Identify the incorrect statement:

1.	A person with blood group type A has type A antigens on RBC and does not have type B antigen on RBC.
2.	A person with blood group type B has anti A antibodies in plasma and does not have anti B antibodies in plasma.
3.	A person with type AB blood group has type A antigen and type B antigen on RBC but does not have anti A or anti B antibodies in plasma.
4.	A person with blood group type O can receive blood from all types of blood group namely AB, A, B, and O.

153. Which of the following options includes plants that all belong to the Solanaceae family?

1.	Solanum tuberosum, Petunia hybrida, Atropa belladonna, Solanum melongena
2.	Solanum tuberosum, Lathyrus odoratus, Atropa belladonna, Capsicum annuum
3.	Solanum melongena, Petunia hybrida, Pisum sativum, Datura stramonium
4.	Capsicum annuum, Nicotiana tabacum, Atropa belladonna, Lathyrus odoratus

154. The transverse section shown in the given figure is representative of:



1. Monocot stem
2. Monocot root
3. Dicot stem
4. Dicot root

155. Which of the following does not favour the formation of large quantities of dilute urine?
1. Alcohol
2. Caffeine
3. Renin
4. Atrial-natriuretic factor

156. Which of the following statements is not correct?

1.	Every 100 ml of oxygenated blood can deliver around 5 ml of oxygen to the tissue under normal physiological conditions.
2.	Oxygen gas has most potent effect on the central chemoreceptors and plays most vital role in regulation of respiration
3.	Nearly 70 percent of carbon dioxide is carried as bicarbonate in the blood.
4.	Every 100 ml of deoxygenated blood delivers about 4 ml of carbon dioxide to the alveoli.

157. Glomerular filtration rate (GFR) is
1. 125 mL/day
2. 180 L/day
3. 1.25 L/day
4. 1.8 L/day

158. Typically, drupes develop from:

1.	monocarpellary inferior ovaries and are one seeded.
2.	monocarpellary superior ovaries and are one seeded.
3.	multicarpellary superior ovaries and are many seeded.
4.	multicarpellary inferior ovaries and are many seeded.

159. In which type of phyllotaxy do more than two leaves arise from the same node, forming a circular arrangement around the stem, as seen in Alstonia?

1.	Alternate	2.	Opposite
3.	Whorled	4.	Spiral

160. During summer sleep and winter sleep in frogs, the gaseous exchange takes place:
1. Across the alveoli in the lungs.
2. Through the moist buccal mucosa.
3. In the gills.
4. Through the skin.

161. Consider the given two statements:

Assertion (A):	Glomerular filtration is considered as a process of ultra-filtration.
Reason (R):	The glomerular capillary blood pressure causes filtration through three layers.

1.	**(A)** is True but **(R)** is False.
2.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
3.	**(A)** is False but **(R)** is True.
4.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.

162. Sweet potato is a modified:
1. Stem
2. Adventitious root
3. Taproot
4. Rhizome

163. In pea and bean flowers, there are five petals, the largest (A) overlaps the two lateral petals (B) which in turn overlap the two smallest anterior petals (C); this type of aestivation is known as (D).

	A	B	C	D
1.	Glume	Wings	Keel	valvate
2.	Standard	Wings	Keel	Valvate
3.	Standard	Keel	Wings	Vexillary
4.	Standard	Wings	Keel	Vexillary

164. Match the pulmonary volume/capacity given in Column I with the corresponding description in Column II and select the correct match from the codes given:

	Column I		Column II
A	Expiratory reserve volume	P	Additional volume of air, a person can inspire by a forcible inspiration.
B	Inspiratory reserve volume	Q	Additional volume of air, a person can expire by a forcible expiration.
C	Residual volume	R	Volume of air that will remain in the lungs after a normal expiration.
D	Functional residual capacity	S	Volume of air remaining in the lungs even after a forcible expiration.

Codes:

	A	B	C	D
1.	P	Q	R	S
2.	P	Q	S	R
3.	Q	P	R	S
4.	Q	P	S	R

165. Regarding plants of economic importance in Family Solanaceae:

I:	Ashwagandha is an ornamental plant.
II:	Tobacco has fumigatory properties.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

166. Which of the following is not an anatomical feature of monocot stems?

1.	Collenchymatous hypodermis
2.	A large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath
3.	Vascular bundles are conjoint and closed
4.	Water-containing cavities are present within the vascular bundles

167. In reference to the given figure, the position of ovary and type of flowers respectively will be:

1. Superior, Hypogynous
2. Inferior, Epigynous
3. Half-inferior, Perigynous
4. Inferior, Epigynous

168. What characteristic feature distinguishes the xylem of gymnosperms from that of angiosperms?

1.	Presence of xylem fibres
2.	Presence of tracheids
3.	Absence of vessels
4.	Absence of xylem parenchyma

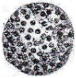
169. In the regulation of respiration, a chemosensitive area adjacent to the rhythm centre in the medulla region of the brain, is highly sensitive to:
1. HCO_3^-
2. CO_2
3. O_2
4. N_2

170. What is the primary factor that determines the direction of gas exchange in the alveoli?

1.	The solubility of gases in water
2.	The thickness of the alveolar membrane
3.	The partial pressure gradient of gases
4.	The hemoglobin concentration in blood

171. Serum differs from blood plasma in:
1. lacking globulins
2. lacking albumins
3. lacking clotting factors
4. lacking antibodies

172. The figure shows a human blood cell. Identify it and give its characteristics:



	Blood cell	Characteristics
1.	Basophils	Secrete serotonin
2.	B-lymphocytes	form about 20% of blood cells involved in immune response
3.	Neutrophils	Most abundant blood cells, phagocytic
4.	Monocytes	Life spans 3 days, produce antibodies

173. Consider the given two statements – an Assertion [A] and a Reason [R] – and select the correct option as your answer:

A:	Monocots do not form secondary tissues.
R:	The vascular bundles have no cambium present in them.

1.	Both Assertion and Reason are true but Reason does not correctly explain Assertion
2.	Assertion is true; Reason is false
3.	Both Assertion and Reason are true and Reason correctly explains Assertion
4.	Assertion is false; Reason is true

174. Consider the following tissues in the stellar region of a stem showing secondary growth.
(A) Primary xylem
(B) Secondary xylem
(C) Primary phloem
(D) Secondary phloem
Arrange these in the correct sequence of their position from pith towards cortex.

1.	(A), (B), (D), (C)	2.	(B), (A), (C), (D)
3.	(A), (B), (C), (D)	4.	(B), (A), (D), (C)

175. What function does the vasa recta play in the counter-current mechanism?

1.	It facilitates the reabsorption of glucose and amino acids.
2.	It exchanges NaCl between its descending and ascending limbs.
3.	It transports water into the medullary interstitium.
4.	It secretes nitrogenous wastes into the nephron.

176. How many of the given statements about aestivation are correct?

I:	Aestivation refers to the arrangement of sepals or petals in a floral bud with respect to the other members of the same whorl.
II:	In valvate aestivation, the sepals or petals in a whorl overlap one another without any specific pattern.
III:	Twisted aestivation is characterized by one margin of the appendage overlapping that of the next one, as seen in china rose and cotton.
IV:	Imbricate aestivation is seen in flowers like pea and bean, where the petals overlap in a specific direction.
V:	Vexillary aestivation, also known as papilionaceous, is found in pea and bean flowers where the largest petal overlaps the lateral petals, which in turn overlap the smallest anterior petals.

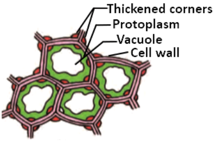
1. 2
2. 3
3. 4
4. 5

177. A right shift in the oxygen - haemoglobin dissociation curve is expected to occur when there is:
1. High pCO_2
2. High pO_2
3. Low pCO_2
4. Less H^+ concentration

178. In an isobilateral (monocotyledonous) leaf, the mesophyll is:

1.	Differentiated into palisade and spongy parenchyma
2.	Composed only of collenchymatous cells
3.	Not differentiated into palisade and spongy parenchyma
4.	Composed of sclerenchyma cells only

179. Identify the simple tissue of plants shown in the given figure:



1. Parenchyma
2. Collenchyma
3. Sclereids
4. Sclerenchyma fibres

180. At the alveoli:

I:	The factors are all favourable for the formation of oxyhemoglobin.
II:	Carbon dioxide trapped as bicarbonate at the tissue level is released out as carbon dioxide.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect